



Marine Engineering Knowledge (MEK)

30 Practice Questions and Answers for STCW Exams

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Practice Set 1

Q1. Boiler is related to _____

- A) Steam Generation
- B) Human waste
- C) Storing Vegetables
- D) Steering

Correct Answer: A (Steam Generation)

Explanation: A boiler is a crucial piece of equipment onboard a ship responsible for generating steam. This steam is utilized for various purposes, including propulsion in steam-powered vessels and for heating systems.

Q2. Which of the following is bottom most part of a diesel engine?

- A) Connecting Rod
- B) Piston
- C) Turbo charger
- D) Exhaust valve

Correct Answer: A (Connecting Rod)

Explanation: The connecting rod is positioned between the piston and the crankshaft. It is the bottom-most component among the choices provided that is directly involved in transmitting the linear motion of the piston to the rotating motion of the crankshaft.

Q3. Main Engine is part of the _____ system

- A) Ballast
- B) Electrical
- C) Propulsion
- D) Bilge

Correct Answer: C (Propulsion)

Explanation: The main engine is the primary source of power for a ship's movement, providing the necessary force to turn the propeller. Therefore, it is an integral part of the propulsion system.

Q4. Domestic Fridge is related to _____.

- A) Human waste
- B) Steering
- C) Starting of Main Engine
- D) Storing Vegetables

Correct Answer: D (Storing Vegetables)

Explanation: A domestic refrigerator onboard is used for the primary purpose of storing perishable items like vegetables, ensuring



Q5. Which of the following is top most part of a diesel engine?

- A) piston
- B) cylinder head
- C) crank shaft
- D) crankcase door

Correct Answer: B (cylinder head)

Explanation: The cylinder head is the topmost component of a diesel engine cylinder, sealing the combustion chamber and housing the valves and injectors. It sits above the piston and cylinder liner.

Q6. Which of the following is lower most part of a diesel engine?

- A) turbo charger
- B) exhaust valve
- C) bed plate
- D) piston

Correct Answer: C (bed plate)

Explanation: The bed plate forms the base of the diesel engine structure, supporting the crankshaft and carrying the main engine load. It is the lowest structural part of the engine assembly.

Q7. Which of the following is within the engine room?

- A) paint storage room
- B) CO2 Room
- C) cargo pump room
- D) purifiers room

Correct Answer: D (purifiers room)

Explanation: The purifiers room (or centrifuge room) is dedicated to housing the machinery used for purifying fuel and lubricating oils, essential for the operation of the main engine and auxiliary machinery. This room is typically located within the engine room complex.

Q8. Which of the following is not within the engine room?

- A) fresh water generator
- B) fire pump
- C) fridge room
- D) refrigeration machinery

Correct Answer: C (fridge room)

Explanation: The fridge room, used for storing provisions, is typically located in the galley or on deck, not within the engine room where machinery operates. The fresh water generator, fire pump, and refrigeration machinery are all essential engine room equipment.

Q9. Which of the following is not within the engine room?

- A) boiler
- B) mooring winch
- C) air compressor
- D) generator engine

Correct Answer: B (mooring winch)



Q10. Which of the following is not a job of an engine room rating?

- A) assist in watchkeeping
- B) assist in watchkeeping
- C) assist in maintenance work
- D) operate main engine

Correct Answer: D (operate main engine)

Explanation: Operating the main engine is a responsibility typically held by the engineering officer of the watch or a senior engineer, not an engine room rating. Ratings assist in watchkeeping duties, perform maintenance tasks, and execute orders from their superiors.

Q11. Which of the following is job of an engine room rating?

- A) start and stop engine
- B) overhaul air compressor
- C) assist in maintenance work
- D) report to bridge about engine condition

Correct Answer: C (assist in maintenance work)

Explanation: An engine room rating's primary duties include assisting in watchkeeping, performing routine maintenance under supervision, and carrying out specific tasks assigned by engineers. Starting and stopping the main engine, overhauling components, or reporting directly to the bridge are generally not their responsibilities.

Q12. Boiler is related to _____.

- A) Storing Vegetables
- B) Human waste
- C) Steam Generation
- D) Steering

Correct Answer: C (Steam Generation)

Explanation: A boiler's primary function is to generate steam, which can be used for various purposes on board, such as heating, driving machinery, or powering auxiliary equipment. It is not involved in storing vegetables, handling human waste, or steering the vessel.

Q13. Sewage Treatment Plant is related to _____.

- A) Starting of Main Engine
- B) Steering
- C) Human waste
- D) Storing Vegetables

Correct Answer: C (Human waste)

Explanation: The Sewage Treatment Plant (STP) is specifically designed to process human waste generated by the crew and passengers, treating it to meet environmental discharge standards as per MARPOL regulations. It is unrelated to engine starting, steering, or food storage.



- A) Boiler
- B) Purifier
- C) Air compressor
- D) Main Engine

Correct Answer: D (Main Engine)

Explanation: The main engine is a complete functional unit that propels the ship. The boiler, purifier, and air compressor are individual systems or machines that are often part of the ship's overall engineering setup, but the main engine itself is the primary propulsion machine.

Q15. A "BOLT" has

- A) Hexagonal head and threads at other end
- B) Thread at both ends
- C) Slotted head and threads at other end
- D) Double pitch threads

Correct Answer: A (Hexagonal head and threads at other end)

Explanation: A bolt is a fastener characterized by external screw threads and typically a hexagonal head, designed to be inserted through unthreaded holes in assembled parts and secured by a nut. It is not a threaded rod or a screw with a slotted head.

Q16. "Cuts" are caused by

- A) Flat spanners
- B) Blunt chisels
- C) Hammers
- D) Sharp objects

Correct Answer: D (Sharp objects)

Explanation: Cuts, abrasions, and lacerations are typically caused by sharp objects such as knives, burrs on metal, or broken glass. While hammers and blunt chisels can cause blunt force trauma, they are not the direct cause of cuts.

Q17. A DIE can be is used to make threads on a pipe

- A) True
- B) False

Correct Answer: A (True)

Explanation: A die is a cutting tool used to create external screw threads on pipes or rods. Therefore, it is indeed used to make threads on a pipe, typically for connections or fittings.

Q18. For lifting a heavy load, you will use

- A) Eye bolt
- B) Air Cylinder
- C) Hydraulic Jack
- D) Chain Block

Correct Answer: D (Chain Block)

Explanation: A chain block is a mechanical device used for lifting heavy loads, consisting of a chain and pulley system. While an eye bolt can be used for lifting, it's a component, and a hydraulic jack lifts by fluid pressure; a chain block is the general tool for heavy lifting.



- A) True
- B) False

Correct Answer: A (True)

Explanation: A tap is a tool used to create or repair internal screw threads, typically by being inserted into a pre-drilled hole and rotated. It is the counterpart to a die, which creates external threads.

Q20. A "STUD" has

- A) Threads at one end with a slot at the other end
- B) Double pitch threads
- C) Hexagonal head and thread at the other end
- D) Thread at both ends

Correct Answer: D (Thread at both ends)

Explanation: A stud is a threaded fastener that has threads on both ends and no head, used for connecting two components where nuts are typically threaded onto both ends. It differs from a bolt, which has a head and threads at one end for a nut.

Practice Set 2

Q1. Which of this is not fitted on the cylinder head of a diesel engine

- A) exhaust valve
- B) turbo charger
- C) starting valve
- D) indicator valve

Correct Answer: B (turbo charger)

Explanation: A turbocharger is an exhaust gas-driven turbine that forces more air into the engine cylinders to increase power and efficiency. It is not a component directly fitted on the cylinder head itself, unlike the exhaust valve, starting valve, and indicator valve.

Q2. Which of this is not a moving part of a diesel engine

- A) piston
- B) crankshaft
- C) crankcase
- D) connecting rod

Correct Answer: C (crankcase)

Explanation: The crankcase is the housing for the crankshaft and part of the engine block; it is a stationary component. The piston, crankshaft, and connecting rod are all essential moving parts within the diesel engine's power cycle.

Q3. Generator engine on board ship is started using

- A) hydraulic oil motors/pressure
- B) compressed air
- C) turbo generator
- D) air compressor

Correct Answer: B (compressed air)

Explanation: Generator engines on ships are typically started using compressed air stored in receiver tanks, which engages the engine's starting mechanism. Hydraulic systems or air compressors are used to supply this air, but the immediate starting medium is



Q4. Purpose of fresh water circulation through diesel engine cylinder is to

- A) prevent corrosion
- B) remove heat of combustion
- C) clean exhaust gases
- D) lubricate moving part

Correct Answer: B (remove heat of combustion)

Explanation: The primary purpose of circulating fresh water through a diesel engine's cooling system is to absorb and remove the immense heat generated by combustion, thereby maintaining optimal operating temperatures. It does not primarily prevent corrosion or clean exhaust gases.

Q5. Turning gear on main engine is provided for

- A) releasing parts if jammed
- B) turning the engine during overhaul
- C) remove exhaust gases from cylinder
- D) ensure turning of propeller

Correct Answer: B (turning the engine during overhaul)

Explanation: The turning gear is provided on the main engine to allow slow rotation of the crankshaft during maintenance, overhaul, or pre-starting checks. This ensures even cooling, allows access to components, and prevents seizure.

Q6. Exhaust gas economiser produces

- A) fresh water
- B) steam
- C) inert gas
- D) electricity

Correct Answer: B (steam)

Explanation: An exhaust gas economizer is a heat exchanger that recovers waste heat from the exhaust gases of the main engine or auxiliary boilers to generate steam or preheat feedwater. It is not involved in producing fresh water, inert gas, or electricity directly.

Q7. Turbo chargers on diesel engine

- A) acts as fuel charger
- B) decreases the pressure of intake air
- C) increases pressure of intake air
- D) increases RPM

Correct Answer: C (increases pressure of intake air)

Explanation: Turbochargers are used to increase the power output of diesel engines by compressing and forcing more intake air into the cylinders. This results in higher volumetric efficiency and improved combustion, not a decrease in air pressure.

Q8. Exhaust gases from the main engine go to the atmosphere through

- A) inert gas generator
- B) oil fired boiler
- C) exhaust fan
- D) exhaust gas boiler

Correct Answer: D (exhaust gas boiler)

Explanation: Exhaust gases from the main engine, after potentially passing through an exhaust gas boiler (economizer) for heat



Q9. Relief valves are provided on main engine cylinders for protection against high

- A) pressure
- B) exhaust temperature
- C) speed
- D) level of water

Correct Answer: A (pressure)

Explanation: Cylinder head relief valves are safety devices designed to protect the engine from potentially catastrophic over-pressurization within the combustion chamber. This can occur due to excessive fuel injection or other abnormal combustion events, not high exhaust temperature or speed.

Q10. Function of a cylinder head relief valve is to

- A) prevent crankcase explosion
- B) stop the engine
- C) reduce the speed of engine
- D) release excessive pressure

Correct Answer: D (release excessive pressure)

Explanation: A cylinder head relief valve is a safety device that opens automatically to release excessive pressure from within the cylinder during operation. This prevents damage to the engine components and potential hazards from over-pressurization.